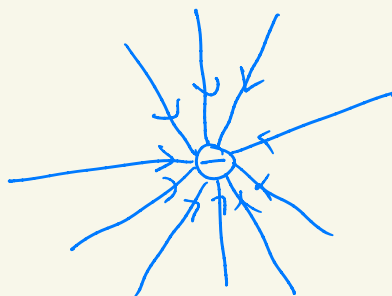
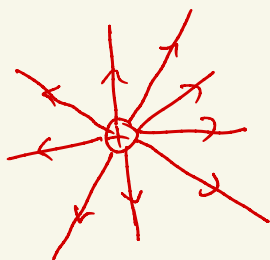


Ch 21: Electric charge
Ch 22: Electric fields

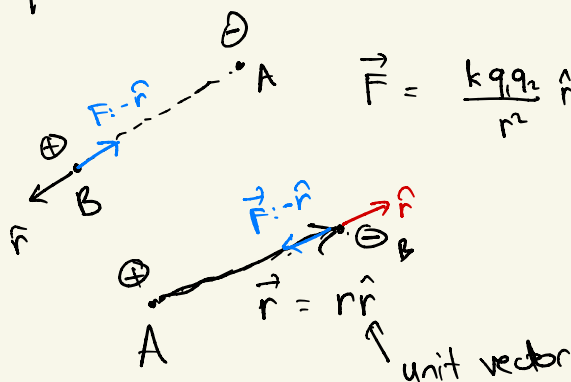
$$\vec{F} = \frac{k q_1 q_2}{r^2} \hat{r}$$



Elementary charge: $+e$ proton, $m_p = 1.67 \times 10^{-27} \text{ kg}$, $938.27 \frac{\text{MeV}}{c^2}$
 $-e$ electron, $m_e = 9.1 \times 10^{-31} \text{ kg}$, $0.511 \frac{\text{MeV}}{c^2}$

proton consists of $2u + d$ quarks
 $\begin{matrix} +\frac{2}{3}e & & +\frac{2}{3}e \\ & u & u \\ & d & -\frac{1}{3}e \end{matrix} \rightarrow +e$

Experiment of e-mass moment was done in 1909 by Rob. Millikan (1923 Nobel Prize)



$$\vec{F} = \frac{k q_1 q_2}{r^2} \hat{r}, \quad k = \frac{1}{4\pi\epsilon_0}, \quad \frac{1}{4\pi\epsilon}$$

ϵ : permittivity of free space

$q_1: +, q_2: - \quad F: -\hat{r} \quad \underline{\text{attracts}}$

$q_1: -, q_2: - \mid q_1: +, q_2: + \quad F: +\hat{r} \quad \underline{\text{repels}}$

Гравитационная сила

$$F_G = -\frac{Gm_1m_2}{r^2} \hat{r}$$

Электр. сила

$$F_E = k \frac{q_1q_2}{r^2} \hat{r}$$

$$G = 6.67 \times 10^{-11} \text{ kg}^{-1} \text{ m}^3 \text{ s}^{-2}$$

$$N = [\text{kg} \cdot \text{m} \cdot \text{s}^{-2}] \text{ (ma)}$$

$$k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2}$$

$$G \cdot \frac{\text{kg}^2}{\text{m}^2} = \text{kg} \cdot \text{m} \cdot \text{s}^{-2}$$

G: k 20 orders of magnitude difference

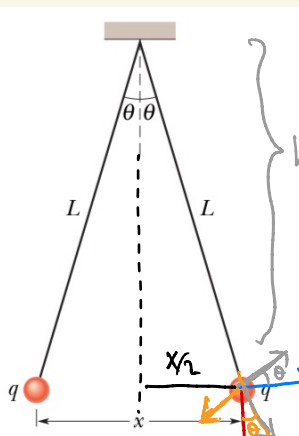
$$[G] = \text{kg}^{-1} \cdot \text{m}^3 \cdot \text{s}^{-2}$$

$F_E \gg F_G$, F_G only comparable at super-massive scale

42 In Fig. 21-38, two tiny conducting balls of identical mass m and identical charge q hang from nonconducting threads of length L . Assume that θ is so small that $\tan \theta$ can be replaced by its approximate equal, $\sin \theta$. (a) Show that

$$x = \left(\frac{q^2 L}{2\pi\epsilon_0 mg} \right)^{1/3}$$

gives the equilibrium separation x of the balls. (b) If $L = 120$ cm, $m = 10$ g, and $x = 5.0$ cm, what is $|q|$?



$$\sin \theta = \frac{x/2}{L}$$

$$\cos \theta = \frac{h}{L}$$

$$\frac{\sin \theta}{\cos \theta} = \tan \theta = \frac{x/2}{h}$$

$$\frac{1}{4\pi\epsilon_0} \frac{q^2}{x^2} = F_E$$

Method 1:

это загрузка по Torque

$$\vec{\tau}_{\text{net}} = 0$$

$$\vec{\tau} = \vec{r} \times \vec{F} = rF \sin \theta$$

$$-mg \sin \theta + \frac{1}{4\pi\epsilon_0} \frac{q^2}{x^2} \cos \theta = 0$$

$$\frac{1}{4\pi\epsilon_0} \frac{q^2}{x^2} \cos\theta = \sin\theta \, mg$$

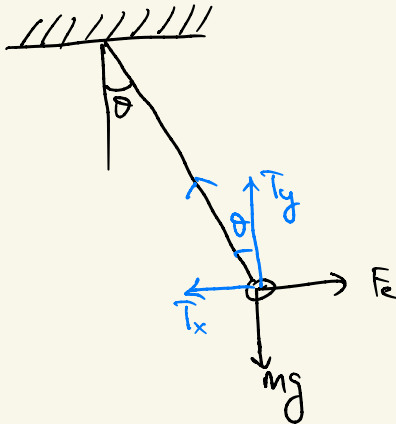
$$\frac{\sin\theta}{\cos\theta} = \frac{q^2}{4\pi\epsilon_0 mg x^2} \approx \sin\theta = \frac{x/L}{L}$$

small angles $\tan\theta \approx \sin\theta$ notomy 20 $\cos\theta \rightarrow 1$

$$^3\sqrt{x^3} = ^3\sqrt{\frac{2q^2L}{4\pi\epsilon_0 mg}}$$

$$x = \left(\frac{q^2 L}{2\pi\epsilon_0 mg} \right)^{1/3}$$

Method 2: с номоментами (Tension)



$$T_y = T \cdot \cos\theta$$

$$T_x = T \cdot \sin\theta$$

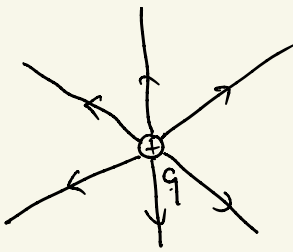
$$T_y = mg$$

$$T_x = F_e$$

$$\frac{T_x}{T_y} = \tan\theta = \frac{F_e}{mg}$$

$$\tan\theta = \frac{\frac{1}{4\pi\epsilon_0} \frac{q^2}{x^2}}{mg} : \tan\theta \approx \sin\theta = \frac{x/L}{L}$$

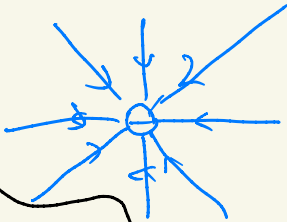
$$x^3 = \frac{q^2 L}{2\pi\epsilon_0 mg}$$



$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} = \frac{kq}{r^2} \hat{r}$$

$$q: +, \quad \vec{E}: +\hat{r}$$

$$q: -, \quad \vec{E}: -\hat{r}$$



Continuous distribution:

Integrate

$$d\vec{E} = k \frac{dq}{r^2} \hat{r}$$

$$\lambda = \frac{Q}{L}$$

$$\sigma = \frac{Q}{A}, \quad \rho = \frac{Q}{V}$$

3 бага поти-ом
запрос

***32 In Fig. 22-50, positive charge $q = 7.81 \text{ pC}$ is spread uniformly along a thin nonconducting rod of length $L = 14.5 \text{ cm}$. What are the (a) magnitude and (b) direction (relative to the positive direction of the x axis) of the electric field produced at point P , at distance $R = 6.00 \text{ cm}$ from the rod along its perpendicular bisector?

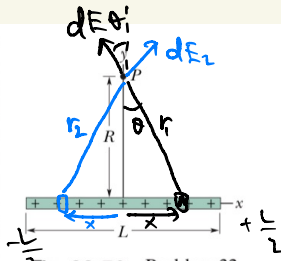


Fig. 22-50 Problem 32.

***33 In Fig. 22-51, a "semi-infinite" nonconducting rod (that is, infinite in one direction only) has uniform linear charge density λ . Show that the electric field $\vec{E}_p$ at point P makes an angle of 45° with the rod and that this result is independent of the distance R . (Hint: Separately find the component of $\vec{E}_p$ parallel to the rod and the component perpendicular to the rod.)

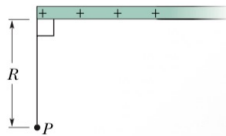


Fig. 22-51 Problem 33.

Pr. 32, Ch 22

$$d\vec{E} = \frac{k dq}{r^2}$$

как мы будем E_x рассчитывать

нам нужно будет только вычислить E_x
горизонтальную.

$$\vec{E}_p = \int d\vec{E}_y, \quad \vec{E}_p = \int dE_x$$

$$= \int_{-L/2}^{+L/2} \frac{k dq}{r^2} \sin\theta \cos\theta$$

$$E_{x, \text{tot}} = \int_{-L/2}^{+L/2} \frac{k dq}{(x^2 + R^2)} \frac{x}{\sqrt{x^2 + R^2}}$$

$$= \int_{-L/2}^{+L/2} \frac{k \lambda x dx}{(x^2 + R^2)^{3/2}}$$

$$\text{Здесь } \sin\theta = \frac{x}{\sqrt{R^2 + x^2}}, \quad \frac{R}{\sqrt{R^2 + x^2}} = \cos\theta$$

$$r = \sqrt{x^2 + R^2}$$

$$dq = \lambda dx = \frac{q}{L} dx$$

$$\vec{E}_{\text{tot}, x} = \int dE_x = k\lambda \int_{-\frac{L}{2}}^{+\frac{L}{2}} \frac{x dx}{(x^2 + R^2)^{3/2}}$$

Method of Substitution: let $u = x^2 + R^2$

then $\frac{du}{dx} = 2x \rightarrow x dx = \frac{du}{2}$

$$= k\lambda \int_{u_i}^{u_f} \frac{du/2}{u^{3/2}}$$

$$u_i = -\frac{L}{2} \quad \times$$

$$\begin{aligned} u_i &= \left(-\frac{L}{2}\right)^2 + R^2 \\ u_f &= \left(+\frac{L}{2}\right)^2 + R^2 \end{aligned}$$

$$= \frac{k\lambda}{2} \int_{u_i}^{u_f} u^{-3/2} du$$

remember $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

$$= \frac{k\lambda}{2} \frac{1}{-\frac{1}{2}} u^{-1/2} \Big|_{u_i}^{u_f}$$

$$= -k\lambda \sqrt{u} \Big|_{u_i}^{u_f}$$

put u back

$$= -k\lambda \sqrt{x^2 + R^2} \Big|_{-\frac{L}{2}}^{+\frac{L}{2}}$$

$$= \left(-k\lambda \sqrt{\frac{L^2}{4} + R^2}\right) - \left(-\frac{k\lambda}{\sqrt{\frac{L^2}{4} + R^2}}\right)$$

$$\boxed{E_{x, \text{tot.}} = 0}$$

$$E_{\text{tot.}} = \int dE_y = \int_{-\frac{L}{2}}^{+\frac{L}{2}} \frac{k\lambda dx}{(R^2 + x^2)} \frac{R}{(R^2 + x^2)^{1/2}} \cos\theta$$

$$E_{tot.} = \int dE_y = \int_{-\frac{L}{2}}^{+\frac{L}{2}} \frac{k\lambda dx}{(R^2+x^2)} \left(\frac{R}{(R^2+x^2)^{1/2}} \right) \cos\theta$$

Here, Method of Substitution DOESN'T WORK.

$$\int \frac{dx}{(R^2+x^2)^{3/2}} = \frac{x}{R^2(x^2+R^2)^{1/2}} + C$$

Just checked from the Resnick Appendix Table of Int.

$$E_{tot.} = \int dE_y = k\lambda R \left(\frac{x}{R^2(x^2+R^2)^{1/2}} \right) \Big|_{-\frac{L}{2}}^{+\frac{L}{2}}$$

$$= \frac{k\lambda}{\sqrt{\frac{L^2}{4} + R^2}} \underbrace{\left(+\frac{L}{2} - \left(-\frac{L}{2}\right) \right)}_L$$

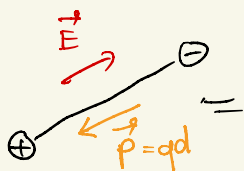
$$E_{tot.} = \frac{k\lambda L}{\sqrt{\frac{L^2}{4} + R^2}} \text{ in } +y \text{ direction}$$

- Принцип суперпозиции: векторное суммирование

$$\vec{F}_{\text{tot}} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots$$

Similar to $\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \dots$

- Диполь (Dipoles)



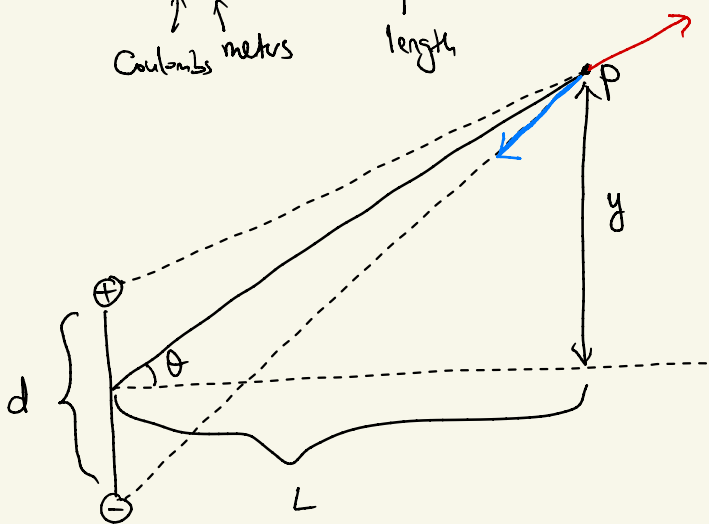
$\vec{p} = qd$ (от минуса к положительному)

$$[p] = [C \cdot m] = [C][L]$$

$\uparrow$ $\uparrow$ $\uparrow$
 Coulombs meters length

$$\vec{E}_p = ?$$

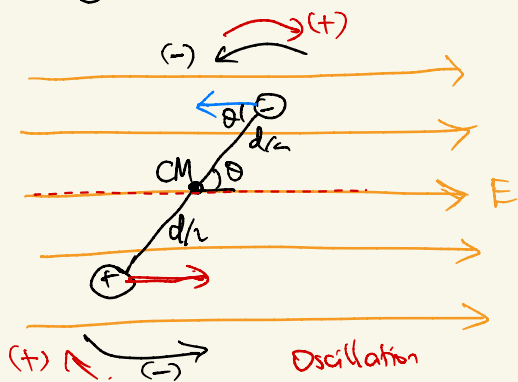
$$d \ll L$$



$$\vec{F} = q\vec{E}$$

$q: +$, $F_{\text{felt by charge}} : +\vec{E}$ - direction

$q: -$, $F_{\text{felt by charge}} : -\vec{E}$ direction



$$\begin{aligned}
 \vec{\tau} &= \vec{r} \times \vec{F} \\
 &= \frac{d}{2} q E \sin \theta + \frac{d}{2} q E \sin \theta \\
 &= q d E \sin \theta = p E \sin \theta = \vec{p} \times \vec{E}
 \end{aligned}$$

Если вас спросят найти расстояние между $\vec{E}$ и $\vec{E}$ как вы найдёте?

$$\vec{\tau} = \vec{r} \times \vec{E}$$

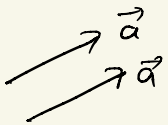
$$\vec{\tau} = I \alpha$$

Just review

$$\vec{L} = \vec{r} \times \vec{p} = m \vec{r} \times \vec{v}, \text{ вращательное движение}$$

$$\vec{L} = I \omega$$

$$\vec{\tau} = \frac{d\vec{L}}{dt} = \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times \frac{d\vec{p}}{dt}$$



$$\vec{a} \times \vec{a} = ? = a \cdot a \cdot \sin \theta$$

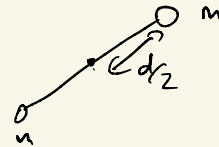
$$= a^2 \sin \theta$$

$$= 0,$$

$$= \vec{v} \times m \vec{v} + \vec{r} \times \vec{F}$$

$$= 0$$

$$\vec{\tau} = \vec{r} \times \vec{F}$$



$$- p E \sin \theta = I \ddot{\theta}$$

when $\sin \theta \approx \theta$:

$$\ddot{\theta} + \frac{p E}{I} \theta = 0$$

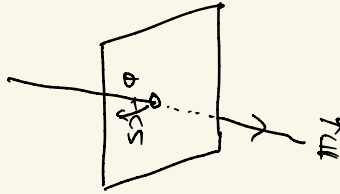
$$I = m \left(\frac{d}{2} \right)^2 + m \left(\frac{d}{2} \right)^2$$

Ch 23: Gauss's Law

скалярная величина

Flux = поток

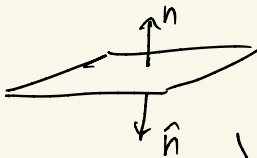
Поток E-поля



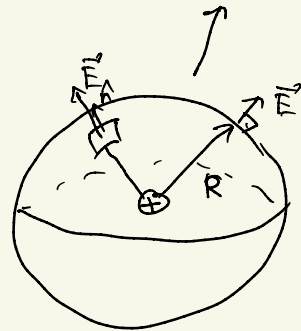
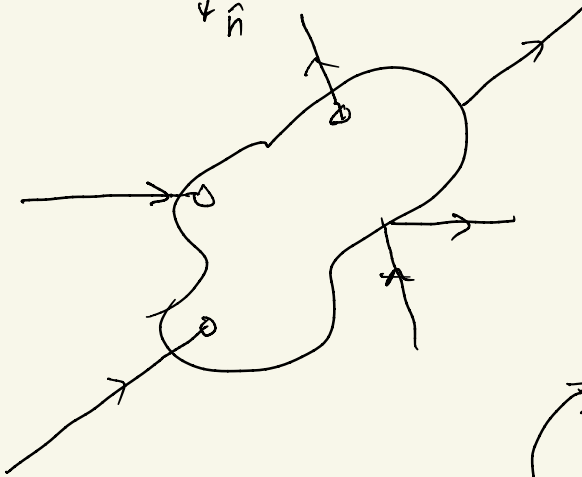
$$\Phi = \vec{E} \cdot \vec{A}$$

$$= |\vec{E}| |\vec{A}| \cos \theta$$

normal vector of any surface is always 90° to it.



Gauss's Law: $\oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{inside}}}{\epsilon_0}$



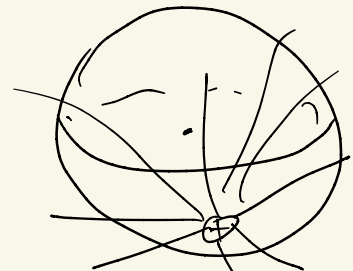
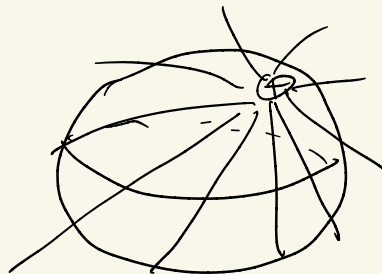
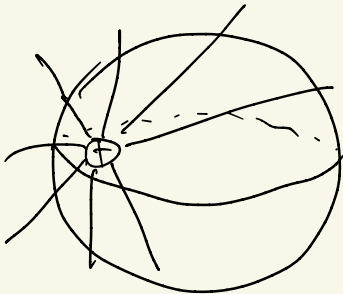
$$E(R) = \frac{q}{4\pi\epsilon_0 R^2}$$

= const

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{in}}}{\epsilon_0}$$

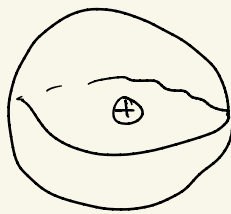
$$E(R) (4\pi R^2) = \frac{q_{\text{ins}}}{\epsilon_0}$$

$$\vec{E}(R) = \frac{q}{4\pi\epsilon_0 R^2} \hat{n}$$



Поток электрического поля не имеет значения

spherical symmetry

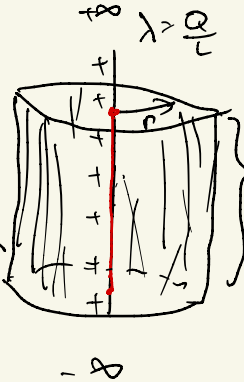
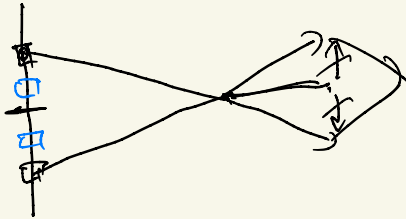


$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

$$E 4\pi r^2 = \frac{q}{\epsilon_0}$$

$$E(r) = \frac{q}{4\pi \epsilon_0 r^2} \hat{r}$$

cylindrical symmetry



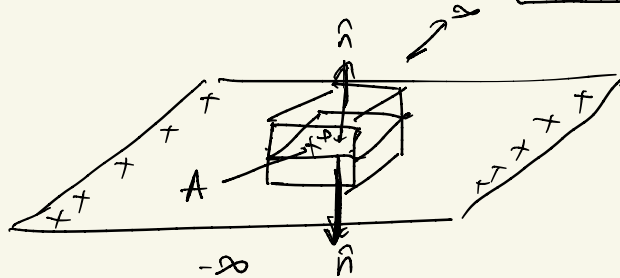
$$E(r) = \text{uniform}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0} = \frac{\lambda l}{\epsilon_0}$$

$$E(r) (2\pi r l) = \frac{\lambda l}{\epsilon_0}$$

$$E(r) = \frac{\lambda}{2\pi \epsilon_0 r} \hat{r}$$

Plane symmetry
(Gaussian pillbox)



в конденсаторах

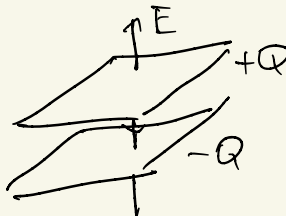
$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

$$E A_{top} + E A_{bottom} = \frac{\sigma A}{\epsilon_0}$$

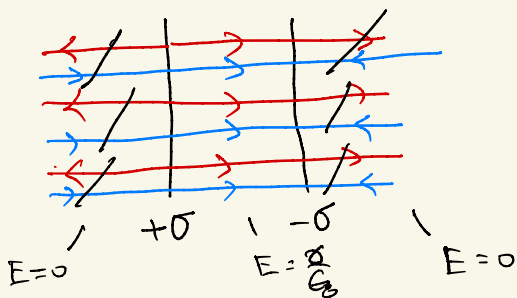
$$\star \quad E = \frac{\sigma}{2\epsilon_0}$$

$\vec{E}$ -field не зависит от расстояния

ex: в конденсаторах



Parallel Plate capacitor



В случае конденсатора с параллельными пластинами

$$E_{\text{tot.}} = \frac{Q}{2\epsilon_0} + \frac{Q}{2\epsilon_0} = \frac{Q}{\epsilon_0}$$

$$E = \frac{Q}{2\epsilon_0}$$

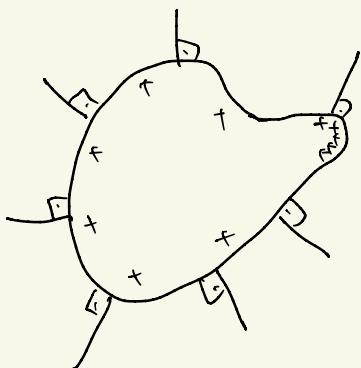
NOTES:

- conductors: they can't be uniformly distributed
- insulators/dielectrics

uniformly distr. charge $\rho = \frac{Q}{\frac{4}{3}\pi R^3} \rightarrow$ not a conductor

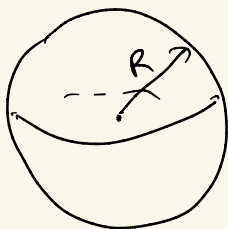


Свойства проводников:



- 1) $\vec{E}_{\text{inside}} = 0$
- 2) $\rho_{\text{inside a conductor}} = 0$
- 3) Any net charge resides on the surface
- 4) Conductor is an equipotential

Ex: Uniformly distributed $\vartheta = \frac{\alpha}{\frac{4\pi}{3}\rho^3}$, find $\mathbb{E}(r|P)$, $\mathbb{E}^2(r|P) = ?$



Hint: Gauss' law

$$\oint \vec{E}(r) \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

for inside the sphere : $r < R$

$$E(r) 4\pi r^2 = \frac{\rho \cdot \frac{4}{3}\pi r^3}{\epsilon_0}$$

$$\vec{E}(r) = \frac{Qr}{3\epsilon_0} \hat{r}$$

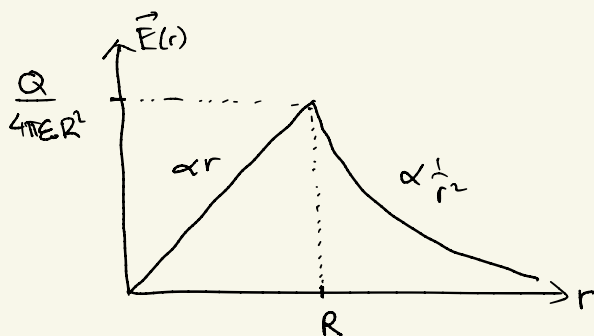
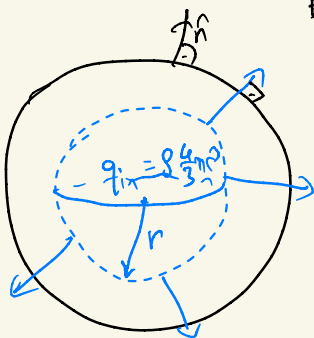
for outside

$$E(r) 4\pi r^2 = \frac{Q}{\epsilon_0}$$

$$\vec{E}(r) = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}$$

$$\vec{E}(r\neq 0) \propto r$$

$$\mathbb{E}(r > R) \propto \frac{1}{r^2}$$



2.5 Conductors

2.5.1 Basic Properties

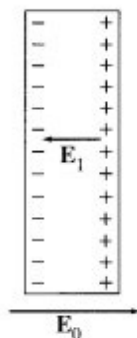
In an **insulator**, such as glass or rubber, each electron is attached to a particular atom. In a metallic **conductor**, by contrast, one or more electrons per atom are free to roam about at will through the material. (In liquid conductors such as salt water it is *ions* that do the moving.) A *perfect* conductor would be a material containing an *unlimited* supply of completely free

2.5. CONDUCTORS

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charges. In real life there are no perfect conductors, but many substances come amazingly close. From this definition the basic electrostatic properties of ideal conductors immediately follow:

(i) **$E = 0$ inside a conductor.** Why? Because if there *were* any field, those free charges would move, and it wouldn't be *electrostatics* any more. Well . . . that's hardly a satisfactory explanation; maybe all it proves is that you can't have electrostatics when conductors are present. We had better examine what happens when you put a conductor into an external electric field E_0 (Fig. 2.42). Initially, this will drive any free positive charges to the right, and negative ones to the left. (In practice it's only the negative charges—electrons—that do the moving, but when they depart the right side is left with a net positive charge—the stationary nuclei—so it doesn't really matter which charges move; the effect is the same.) When they come to the edge of the material, the charges pile up: plus on the right side, minus on the left. Now, these **induced charges** produce a field of their own, E_1 , which, as you can see from the figure, is in the *opposite direction* to E_0 . That's the crucial point, for it means that the field of the induced charges *tends to cancel off the original field*. Charge will continue to flow until this cancellation is complete, and the resultant field inside the conductor is precisely zero.⁷ The whole process is practically instantaneous.



(ii) $\rho = 0$ inside a conductor. This follows from Gauss's law: $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$. If $\mathbf{E} = 0$, so also is ρ . There is still charge around, but exactly as much plus charge as minus, so the net charge density in the interior is zero.

(iii) Any net charge resides on the surface. That's the only other place it *can* be.

(iv) A conductor is an equipotential. For if $\mathbf{a}$ and $\mathbf{b}$ are any two points within (or at the surface of) a given conductor, $V(\mathbf{b}) - V(\mathbf{a}) = -\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\mathbf{l} = 0$, and hence $V(\mathbf{a}) = V(\mathbf{b})$.

⁷ Outside the conductor the field is *not* zero, for here $\mathbf{E}_0$ and $\mathbf{E}_1$ do not cancel.

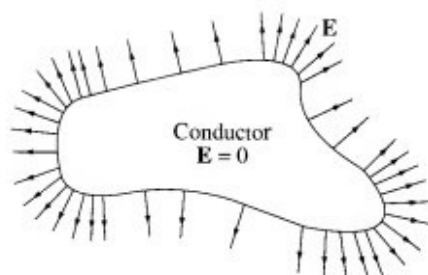


Figure 2.43

(v) $\mathbf{E}$ is perpendicular to the surface, just outside a conductor. Otherwise, as in (i), charge will immediately flow around the surface until it kills off the tangential component (Fig. 2.43). (*Perpendicular* to the surface, charge cannot flow, of course, since it is confined to the conducting object.)

I think it is strange that the charge on a conductor flows to the surface. Because of their mutual repulsion, the charges naturally spread out as much as possible, but for *all* of them to go to the surface seems like a waste of the interior space. Surely we could do better, from the point of view of making each charge as far as possible from its neighbors, to sprinkle *some* of them throughout the volume. . . Well, it simply is not so. You do best to put *all* the charge on the surface, and this is true regardless of the size or shape of the conductor.⁸

The problem can also be phrased in terms of energy. Like any other free dynamical system, the charge on a conductor will seek the configuration that minimizes its potential energy. What property (iii) asserts is that the electrostatic energy of a solid object (with specified shape and total charge) is a minimum when that charge is spread over the surface. For instance, the energy of a sphere is $(1/8\pi\epsilon_0)(q^2/R)$ if the charge is uniformly distributed over the surface, as we found in Ex. 2.8, but it is greater, $(3/20\pi\epsilon_0)(q^2/R)$, if the charge is uniformly distributed throughout the volume (Prob. 2.32).

2.5.2 Induced Charges

If you hold a charge $+q$ near an uncharged conductor (Fig. 2.44), the two will attract one another. The reason for this is that q will pull minus charges over to the near side and repel plus charges to the far side. (Another way to think of it is that the charge moves around in such a way as to cancel off the field of q for points inside the conductor, where the total field must be zero.) Since the negative induced charge is closer to q , there is a net force of attraction. (In Chapter 3 we shall calculate this force explicitly, for the case of a spherical conductor.)

⁸By the way, the one- and two-dimensional analogs are quite different: The charge on a conducting *disk* does not all go to the perimeter (R. Friedberg, *Am. J. of Phys.* **61**, 1084 (1993)), nor does the charge on a conducting *needle* go to the ends (D. J. Griffiths and Y. Li, *Am. J. of Phys.* **64**, 706 (1996)). See Prob. 2.52.

2.5. CONDUCTORS

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Figure 2.44

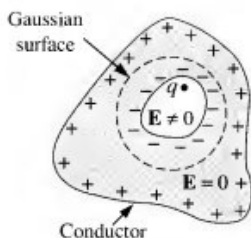


Figure 2.45

By the way, when I speak of the field, charge, or potential “inside” a conductor, I mean in the “meat” of the conductor; if there is some *cavity* in the conductor, and within that cavity there is some charge, then the field *in the cavity* will *not* be zero. But in a remarkable way the cavity and its contents are electrically isolated from the outside world by the surrounding conductor (Fig. 2.45). No external fields penetrate the conductor; they are canceled at the outer surface by the induced charge there. Similarly, the field due to charges within the cavity is killed off, for all exterior points, by the induced charge on the inner surface. (However, the compensating charge left over on the *outer* surface of the conductor effectively “communicates” the presence of q to the outside world, as we shall see in Ex. 2.9.) Incidentally, the total charge induced on the cavity wall is equal and opposite to the charge inside, for if we surround the cavity with a Gaussian surface, all points of which are in the conductor (Fig. 2.45), $\oint \mathbf{E} \cdot d\mathbf{a} = 0$, and hence (by Gauss’s law) the net enclosed charge must be zero. But $Q_{\text{enc}} = q + q_{\text{induced}}$, so $q_{\text{induced}} = -q$.

HW Ch 21

Pc. 7, 22, 23

HW Ch 22

22-7 disk (learn it)

Problems: 9, 18, 19, 24, 25, 27, 29, 30, 31, 33,
53, 72, 84

HW Ch 23

Questions: 1, 9

Problems: 1, 3a, 5, 9, 10, 29, 32, 33, 34, 39, 40, 43,
49, 51, 52, 55, 73, 76
★

Deadline: Thu Aug 14th 12:00 PM noon